

```
import pandas as pd
```

✓ 0.4s

```
insurance = pd.read_csv("insurance.csv")  
insurance.head()
```

✓ 0.0s  Open 'insurance' in Data Wrangler

	age	sex	bmi	children	smoker	region	charges
0	19	female	27.900	0	yes	southwest	16884.92400
1	18	male	33.770	1	no	southeast	1725.55230
2	28	male	33.000	3	no	southeast	4449.46200
3	33	male	22.705	0	no	northwest	21984.47061
4	32	male	28.880	0	no	northwest	3866.85520

```
from autoviz.AutoViz_Class import AutoViz_Class  
AV = AutoViz_Class()  
insurance_report = AV.AutoViz("insurance.csv")
```

[3] ✓ 3.4s

... Imported v0.1.905. Please call AutoViz in this sequence:

```
AV = AutoViz_Class()  
%matplotlib inline  
dfte = AV.AutoViz(filename, sep=',', depVar='', dfte=None, header=0, verbose=1, lowess=False,  
                  chart_format='svg', max_rows_analyzed=150000, max_cols_analyzed=30, save_plot_dir=None)
```

Shape of your Data Set loaded: (1338, 7)

#####

C L A S S I F Y I N G V A R I A B L E S

#####

Classifying variables in data set...

```
Number of Numeric Columns = 2  
Number of Integer-Categorical Columns = 2  
Number of String-Categorical Columns = 1  
Number of Factor-Categorical Columns = 0  
Number of String-Boolean Columns = 2  
Number of Numeric-Boolean Columns = 0  
Number of Discrete String Columns = 0  
Number of NLP String Columns = 0  
Number of Date Time Columns = 0  
Number of ID Columns = 0  
Number of Columns to Delete = 0  
7 Predictors classified...
```

No variables removed since no ID or low-information variables found in data set

To fix these data quality issues in the dataset, import FixDQ from autoviz...

There are 1 duplicate rows in your dataset

Alert: Dropping duplicate rows can sometimes cause your column data types to change to object!

All variables classified into correct types.

	Data Type	Missing Values%	Unique Values%	Minimum Value	Maximum Value	DQ Issue
age	int64	0.000000	3	18.000000	64.000000	No issue
sex	object	0.000000	0			No issue
bmi	float64	0.000000	NA	15.960000	53.130000	Column has 9 outliers greater than upper bound (47.32) or lower than lower bound(13.67). Cap them or remove them.
children	int64	0.000000	0	0.000000	5.000000	No issue
smoker	object	0.000000	0			No issue
region	object	0.000000	0			No issue
charges	float64	0.000000	NA	1121.873900	63770.428010	Column has 139 outliers greater than upper bound (34524.78) or lower than lower bound(-13120.72). Cap them or remove them.

.. Number of All Scatter Plots = 3

All Plots done

Time to run AutoViz = 1 seconds

AUTO VISUALIZATION Completed

```
report = AV.AutoViz("insurance.csv", chart_format="bokeh", depVar="charges")
```

✓ 2.2s

Shape of your Data Set loaded: (1338, 7)

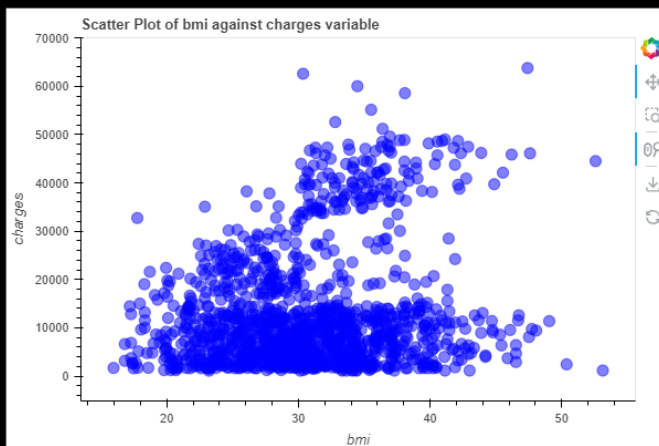
```
#####  
##### CLASSIFYING VARIABLES #####  
#####  
Classifying variables in data set...
```

```
Number of Numeric Columns = 1  
Number of Integer-Categorical Columns = 2  
Number of String-Categorical Columns = 1  
Number of Factor-Categorical Columns = 0  
Number of String-Boolean Columns = 2  
Number of Numeric-Boolean Columns = 0  
Number of Discrete String Columns = 0  
Number of NLP String Columns = 0  
Number of Date Time Columns = 0  
Number of ID Columns = 0  
Number of Columns to Delete = 0  
6 Predictors classified...
```

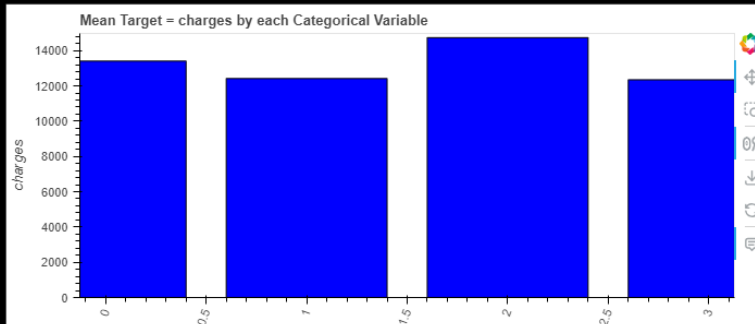
No variables removed since no ID or low-information variables found in data set

```
##### Regression problem #####
```

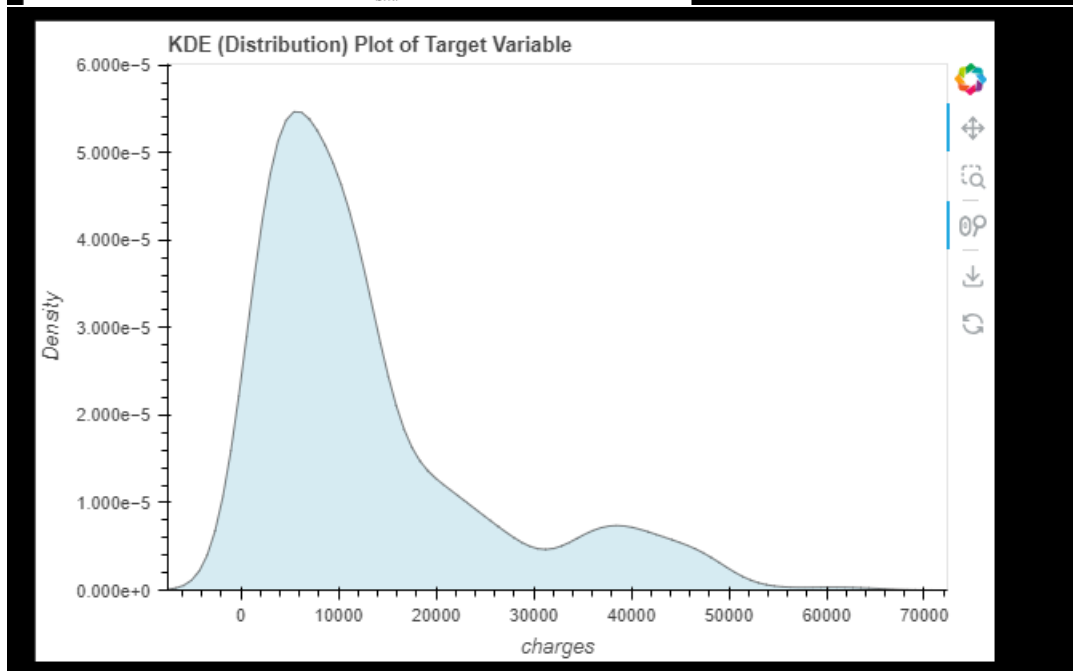
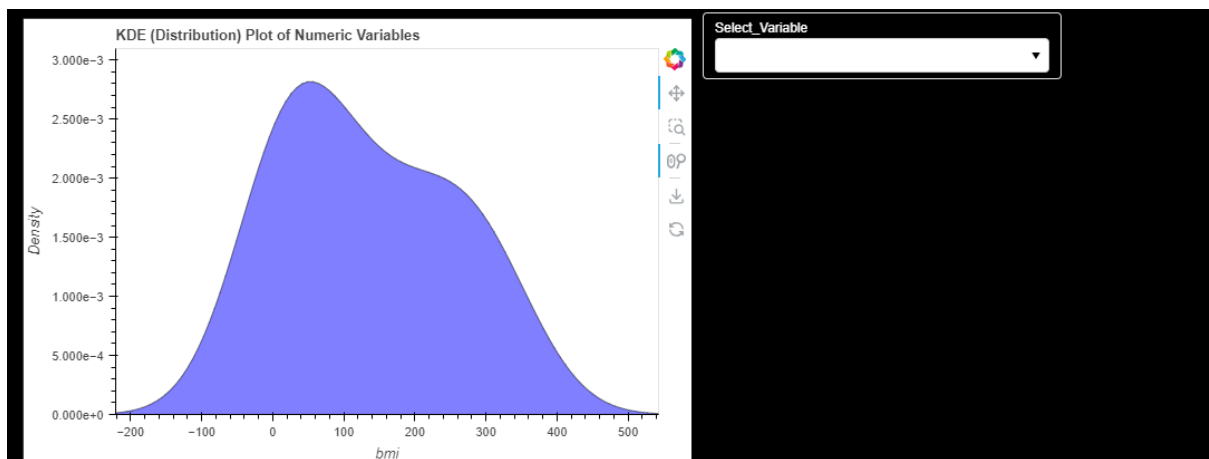
```
##### Regression problem #####
```

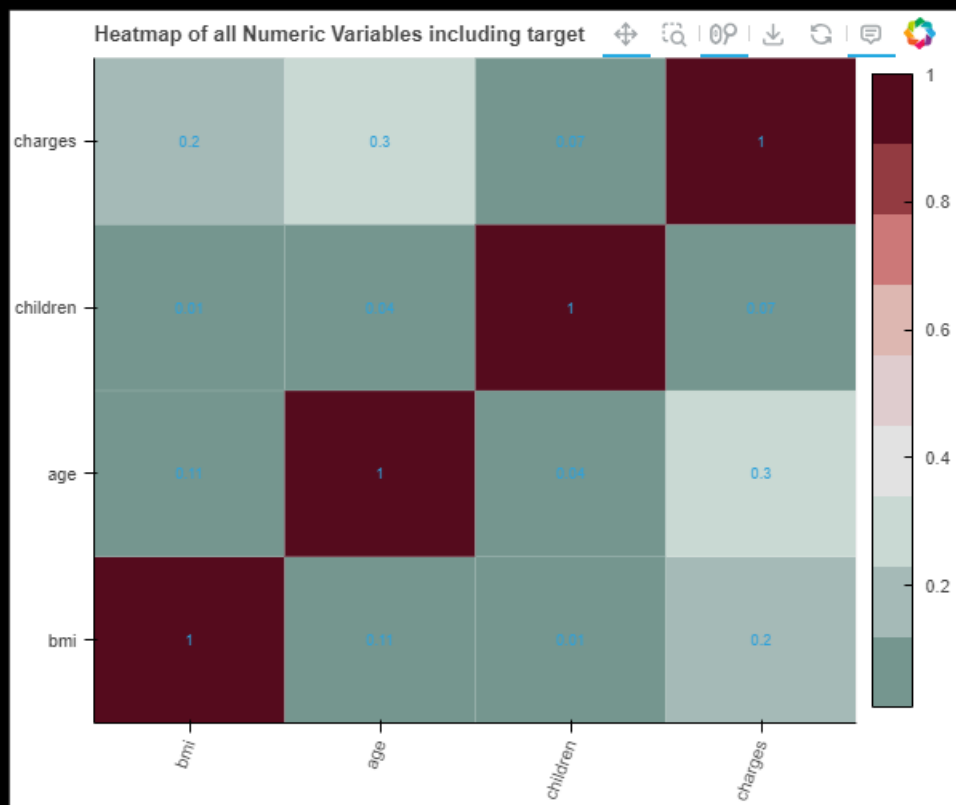


Select_Variable



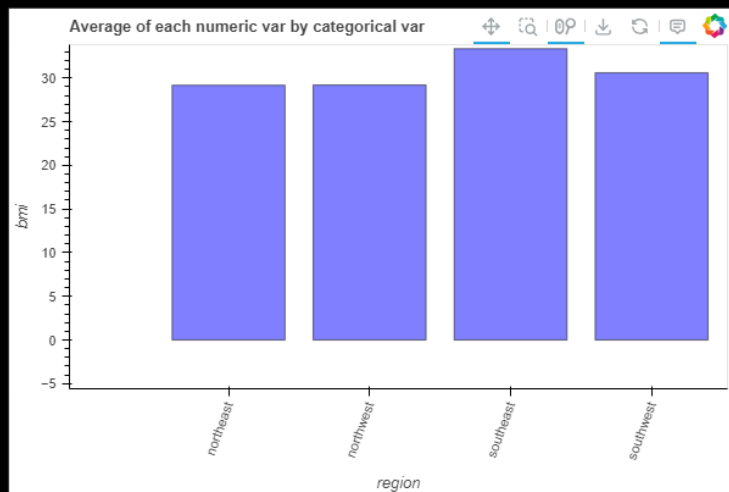
Select_Cat_Variable





X-Axis:

Y-Axis:



Time to run AutoViz (in seconds) = 2